

MORE ON LINEAR REGRESSION & APPLICATIONS

LEAST ANGLE REGRESSION (LAR)

- Extremely efficient algorithm for computing the Lasso path
- At each step we identify the variable that is most correlated with the target. Then, we move it towards its OLS coefficient.

Algorithm

- 1) Normalise predictors. Set $r = y - \bar{y}$, $\beta_1, \beta_2, \dots, \beta_p = 0$
- 2) Find x_j most correlated with r
- 3) Move β_j towards its OLS coefficient until some other x_k has as much correlation with r
- 4) Move β_j, β_k towards OLS until some other x_l has as much correlation with r (now x_l/β_l is in the active set)
- 5) Continue this way until all p predictors have been entered $\min(N-1, p)$ steps

* Let A_k be the active set. At the beginning of the k -th step. Let β_{A_k} be the coefficient vector at that step. Then:

$$r_k = y - X_{A_k} \beta_{A_k} \quad (\text{residual})$$

$$s_k = (X_{A_k}^T X_{A_k})^{-1} X_{A_k}^T r_k \quad (\text{direction of the current step})$$

$$\beta_{A_k}(\alpha) = \beta_{A_k} + \alpha s_k \quad (\alpha: \text{step size})$$

$$c = X^T r \quad (\text{Correlations}) \quad -1 \leq \cos(\alpha) \leq 1$$

$$\cos(0) = 1$$

$$c_j = x_j^T r = \|x_j\| \|r\| \cos \theta_j = \|r\| \cos \theta_j$$

$$\Rightarrow c_j = \|r\| \cos \theta_j \quad (\text{Picking highest } |c_j| \Leftrightarrow \text{Picking small } \theta_j)$$

LEAST ANGLE REGRESSION (Lasso Modification)

Algorithm: If a non-zero coefficient hits zero, drop it from the active set and recompute current joint least squares.

LINEAR REGRESSION IN PRACTICE (CAPM)

$$r_i = \alpha_i + \beta_i r_m + \varepsilon_i$$

r_i : stock/strategy returns

r_m : market returns

ε_i : idiosyncratic noise

α_i : systematic mispricing

$$\hat{\varepsilon}_i = r_i - \hat{\alpha}_i - \hat{\beta}_i r_m \quad (\text{residuals})$$

$$\text{Recall: } \hat{\theta} = \begin{bmatrix} \hat{\alpha}_i \\ \hat{\beta}_i \end{bmatrix} = (X^T X)^{-1} X^T r_i \quad (\text{OLS})$$

$$\text{Var}(\hat{\theta}) = \sigma^2 (X^T X)^{-1} \quad (\widehat{\text{Var}}(\hat{\theta}) = \hat{\sigma}^2 (X^T X)^{-1})$$

$$\hat{\sigma}^2 = \frac{1}{N-2} \hat{\varepsilon}^T \hat{\varepsilon}, \quad \hat{\varepsilon} = y - X \hat{\theta}$$

$$1-\gamma \text{ confidence interval: } \hat{\alpha} \pm z_{1-\gamma/2} \cdot \text{SE}(\hat{\alpha}) \quad (\text{SE}(\hat{\alpha}) = \sqrt{\text{Var}(\hat{\alpha})})$$

$$95\% \text{ CI: } z_{1-\gamma/2} = 1.96\%$$

$$\text{Assuming } \text{Cov}(r_m, \varepsilon_i) = 0$$

$$\text{Var}(A+B) = \text{Var}(A) + \text{Var}(B) + 2\text{Cov}(A, B)$$

$$\text{Var}(r_i) = \text{Var}(\alpha_i + \beta_i r_m + \varepsilon_i)$$

$$= \text{Var}(\alpha_i) + \text{Var}(\beta_i r_m + \varepsilon_i) + 2\text{Cov}(\alpha_i, \beta_i r_m + \varepsilon_i)$$

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])] \quad (\alpha_i \text{ is treated as a const.})$$

$$E[\alpha_i] = \alpha_i \Rightarrow \text{Cov}(\alpha_i, \beta_i r_m + \varepsilon_i) = 0 \quad \text{Also } \text{Var}(\alpha_i) = 0$$

$$\begin{aligned} \Rightarrow \text{Var}(r_i) &= \text{Var}(\beta_i r_m + \varepsilon_i) \\ &= \text{Var}(\beta_i r_m) + \text{Var}(\varepsilon_i) + 2\text{Cov}(\beta_i r_m, \varepsilon_i) \\ &= \beta_i^2 \text{Var}(r_m) + \sigma_{\varepsilon_i}^2 \quad (\text{Var}(cX) = c^2 \text{Var}(X)) \\ &= \text{systematic risk} + \text{idiosyncratic risk} \end{aligned}$$

For your portfolio

$$r_p = w^T \alpha + (w^T \beta) r_m + w^T \varepsilon$$

$$\text{Ideally } w^T \beta = 0 \Rightarrow r_p = w^T \alpha + w^T \varepsilon$$

$$\text{Objective: } \max_w w^T \alpha \quad \text{subject to } w^T \beta = 0$$

Example for two assets:

$$\max_w w_1 \alpha_1 + w_2 \alpha_2 \quad \text{with } w_1 \beta_1 + w_2 \beta_2 = 0$$

$$\Rightarrow \frac{w_1}{w_2} = -\frac{\beta_2}{\beta_1}$$

$$\Rightarrow \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \lambda \begin{bmatrix} \beta_2 \\ -\beta_1 \end{bmatrix} \quad \lambda \in \mathbb{R}$$

$$\therefore w^T \alpha = \lambda (\beta_2 \alpha_1 - \beta_1 \alpha_2)$$

$$\beta_2 \alpha_1 - \beta_1 \alpha_2 > 0 \Rightarrow \lambda > 0$$

$$\beta_2 \alpha_1 - \beta_1 \alpha_2 < 0 \Rightarrow \lambda < 0$$

Exercises

- 1) Compute $\hat{\alpha}, \hat{\beta}$ for a stock of your choice
- 2) Estimate $\text{SE}(\hat{\alpha}), \text{SE}(\hat{\beta})$ and a 95% CI
- 3) Create a β neutral portfolio ($w^T \beta = 0$) by trading the market
- 4) Repeat for other stocks, try and combine 2 or more stocks in your portfolio
- 5) Create a function to compute the Lasso using the LAR algorithm (Hard)